PG – 374

III Semester M.C.A. (Two Years Course) Examination, May 2024
(CBCS) (2021 – 22)
COMPUTER SCIENCE
MCA3E1/MSC3E1 – Machine Learning (Elective)

Time : 3 Hours Max. Marks : 70

Instructions *: 1) Answer* ***any 5*** *from Section –* ***A****.*
2) Answer ***any 4*** *from Section –* ***B****.*

SECTION – A

Answer **any 5** questions. **Each** question carries **6** marks. **(5×6=30)**

1. Expand AUC, RoC and explain the concept.

2. Explain MAP Hypothesis to predict probability.

3. What is Entropy ? How do we employ Mutual Information for classification between positive and negative class ?

4. Define Machine Learning. Mention the perspectives and issues in ML.

5. Explain about the terms :
   i) K-mean clustering
   ii) Hierarchical clustering.

6. Explain Confusion Matrix with respect to detection of “Spam e-mails”.

7. What is an Inductive bias ? Is there any effect on classification due to bias ?

8. In which cases Naive Bayes is useful in classification ? Why ?

P.T.O.

SECTION – B

Answer **any 4** questions. **Each** question carries **10** marks. **(4×10=40)**

9. Explain K-Nearest Neighbour techniques with an example.

10. What is a Recommender System ? How Machine Learning is useful in Recommender Systems ?

11. What is Gibbs Algorithm ? What is its suitability in Machine Learning ?

12. With a suitable example, explain Logistic Regression.

13. Distinguish between supervised learning and reinforcement learning. Illustrate with an example.

14. Write Decision Tree Algorithm and explain with suitable example.

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PG – 673

III Semester M.C.A. (Two Years Course) Examination, May/June 2026
(CBCS) (2021 – 22)
COMPUTER SCIENCE
MCA3E1 : Machine Learning (Elective)

Time : 3 Hours Max. Marks : 70

Instructions : *i) Answer **any five** questions from Part – **A**.*
*ii) Answer **any four** questions from Part – **B**.*

PART – A

Answer **any five** questions. **Each** question carries **6** marks. **(5×6=30)**

1. Summarize the critical issues in Machine Learning.
2. Discuss the steps for designing learning system.
3. Explain K-NN algorithm in detail.
4. Describe Back-propagation algorithm with example.
5. Difference between supervised and unsupervised learning.
6. Explain hidden Markov models with example.
7. Discuss Bayes theorem.
8. Explain Bi-objectives optimization problem using genetic algorithm.

PART – B **(4×10=40)**

9. Discuss the steps for designing learning system with a neat diagram.
10. Explain the importance of artificial neural network.
11. Explain support vector machine with example.
12. a) Differentiate between agglomerartive and divisive clustering. 5
    b) Illustrate apriori algorithm with example. 5
13. a) Explain Frequent Pattern (FP) growth algorithm. 5
    b) Explain metric performance measures for classification. 5
14. Write short on the following.
    a) PCA. 5
    b) Gaussian mixture model. 5

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